

Practice 7

Deadline: Check onsite on or before week 9 lab.

PMD is a source code analyzer or linter. It finds common programming flaws like unused variables, empty catch blocks, unnecessary object creation, and so forth.

You could use it as standalone software, IDE plugin or Maven plugin. We'll use it as a Maven plugin in this practice.

You could add **maven-pmd-plugin** in the **<build>** section of your **pom.xml** to run PMD in your Maven project. By default, PMD checks the project using a default ruleset. In the maven PMD plugin configuration, you can also specify a custom ruleset file to use.

```
<configuration>
  <rulesets>
    <ruleset>customize-rules.xml</ruleset>
  </rulesets>
</configuration>
```

The PMD maven plugin has several goals, including **pmd:pmd**, **pmd:cpd**, **pmd:check**, **pmd:cpd-check**, etc. Please check the documentation for details. You can also specify the goals to execute in the **<executions>** section of the plugin configuration.

```
<executions>
  <execution>
    <goals>
      <goal>check</goal>
    </goals>
  </execution>
</executions>
```

By default, PMD is executed in the **verify** phase of the Maven lifecycle.

✓ com.sismics.docs:docs-corejar:1.12-SNAPSHOT	4 sec, 567 ms
✓ resources	130 ms
✓ compile	213 ms
✓ testResources	5 ms
✓ testCompile	31 ms
✓ test	163 ms
✓ jar	527 ms
✓ pmd	2 sec, 119 ms
✓ check	9 ms
✓ cpd	886 ms
✓ cpd-check	7 ms
✓ install	17 ms

After configuring the PMD plugin in the **pom.xml**, run:

```
mvn clean -DskipTests install
```

Then, you will see pmd report (`pmd.html`) under `docs-core/target/site`, `docs-web/target/site`, and `docs-web-common/target/site`.

Task

In this practice, you'll define a customized PMD ruleset that includes only the following rules that we've covered in the lecture:

- Lines of code (LoC)
- Cyclomatic Complexity (CC)
- Weighted Methods per Class (WMC)
- Coupling Between Objects (CBO)
- Lack of Cohesion in Methods (LCOM)

In addition, we want to include the PMD report in the site documentation of Teedy.

Steps

1. Check [here](#) to learn how to define a customized PMD ruleset.
2. Check [here](#) for a list of available Java Rules. You need to pick the ones that correspond to or are closely related to the above 5 metrics, and add them into your customized ruleset.
3. Configure `pom.xml` in Teedy to use the customized ruleset and generate the PMD report in the site documentation.
4. Run `mvn site`.
5. Open the generated `target/site/index.html` in your browser. The site should now include the PMD report. Note that, the PMD report should include problems detected by the customized ruleset that you defined in step 2.

Docs Parent

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Modules

Docs Core

Docs Web Commons

Docs Web

Project Documentation

Project Information

Project Reports

PMD

CPD

Built by:

PMD Results

The following document contains the results of [PMD](#) 7.0.0.

Violations By Priority

Priority 3

Docs Core - com/sismics/docs/core/dao/GroupDao.java

Rule	Violation
CyclomaticComplexity	The method 'findByCriteria(GroupCriteria, SortCriteria)' has a cyclomatic complexi

Docs Core - com/sismics/docs/core/service/InboxService.java

Rule	Violation
CouplingBetweenObjects	High amount of different objects as members denotes a high coupling
CyclomaticComplexity	The method 'importMessage(Message, Map<String, String>)' has a cyclomatic complexity of 12.